

1. 消除左递归:

$$S \rightarrow (L) \mid a$$

$$\underline{L \rightarrow L, S \mid S} \Rightarrow L \rightarrow S L'$$

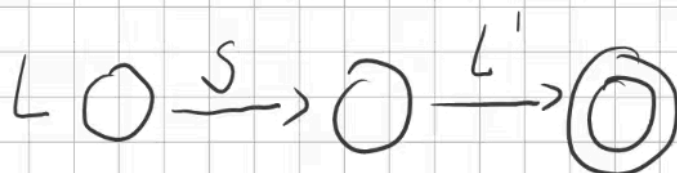
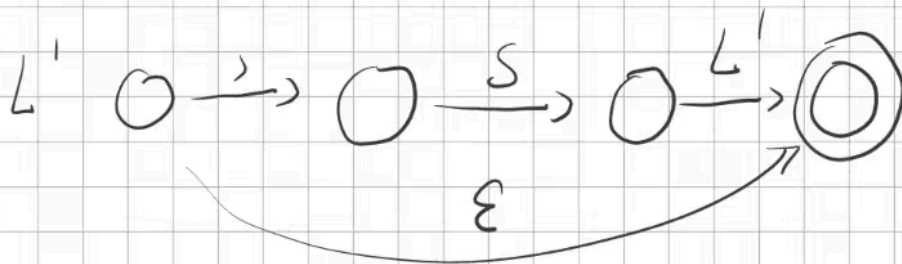
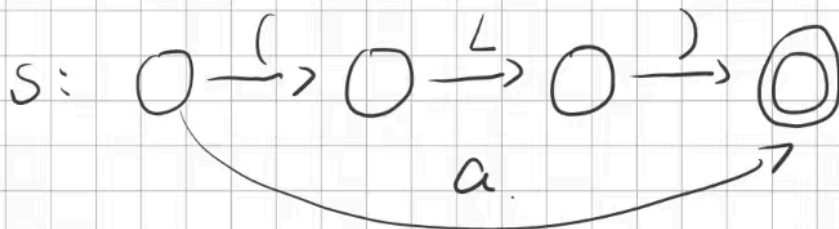
$$\therefore S \rightarrow (L) \mid a$$

$$L \rightarrow S L'$$

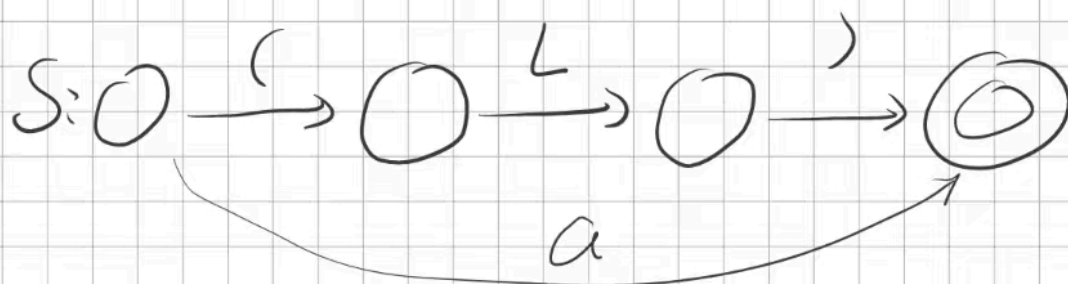
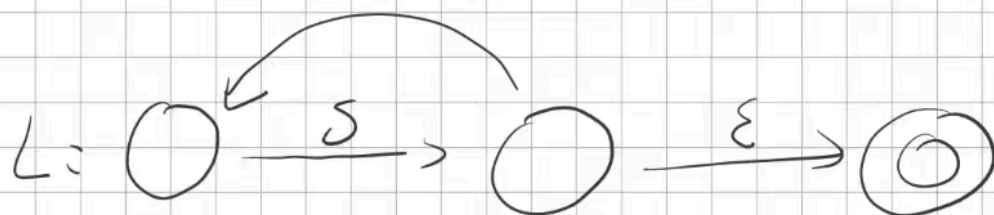
$$L' \rightarrow \epsilon \mid S L'$$

$$L' \rightarrow \epsilon \mid S L'$$

转换图:



化简:



}

```
void L() throws SyntacticException {
```

```
    S();
```

```
    while (lookahead == '>') {
```

```
        match('>');
```

```
        L();
```

```
    }
```

```
}
```

```
void S() throws SyntacticException {
```

```
    if (lookahead == "(") {
```

```
        match("(");
```

```
        L();
```

```
        match(")");
```

```
    } else if (lookahead == 'a') { match(a); }
```

```
    } else { throw SyntacticException; }
```


⑪ 返回⑦中 $L()$

⑫ 返回⑦中 $S()$, 匹配")"

⑬ 返回⑥中 $L()$

⑭ 返回③中 $L()$

⑮ 返回①中 $S()$ 匹配")"退出

匹配结束

因为 $S \rightarrow \varepsilon$ 且

$$FOLLOW(S) = \{ \$, b, a \}$$

$$FIRST(bSaS) = \{ b \}$$

$$FIRST(bSaS) \cap FOLLOW(S)$$

$\neq \emptyset$ 不是 LL(1), 因此也不是
可预测的

$$\text{FIRST}(S) = \{a\}$$

$$\text{FOLLOW}(S) = \{ \$, *, + \}$$